Exploring Dialects within and around Beijing Using DialectR

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Abstract

This study investigates internal variation within Beijing Dialects, a variety of Northern Mandarin. By using dialectometric methods based on phonetic edit distance, we analyze the IPA transcript from 7 sites in Beijing and several nearby sites in Hebei and Tianjin. In this study, we followed the dialectR framework, computing the pairwise dialect distances, visualizing our result based on multidimensional scaling and hierarchical clustering. The results show that while Beijing Dialect exhibits a high level of overall similarity, it still displays internal spatial variation. We compared the Beijing Dialect with nearby dialects site, which also indicate that the internal variation within Beijing Dialects is relative smaller than the variation between Beijing and those nearby external sites. We also found out that the distance similarity does not strictly follow the geographic distance. These findings suggest that the dialects variation within and around Beijing Dialect sites does exist, but is affected by multiple factors.

1 Introduction

Beijing Dialect ("Beijing Hua") is generally regarded as an important variety of Northern Mandarin and is often analyzed within the Mandarin dialect system. Traditionally, Mandarin has been considered to hold relatively limited internal variation compared to non-Mandarin varieties. However, recent studies suggest that Mandarin is more internally diverse than previously assumed, for example, a clear north–south division revealed by NeighborNet analyses (Huang et al., 2024).

Despite all these findings, most of the research on internal variation within Mandarin focuses on large-scale and macro level analyzes. For a single Mandarin variety region, researches on internal differences remain scarce. Therefore, we take Beijing Dialect as the object, use the dialectometry methods, try to examine whether linguistic variation exists within Beijing Dialects region and gain a better understanding of the internal variation structure of it.

Specifically, we discuss the following two research questions:

- **RQ1: Does the Beijing Dialect have the structural internal variation within itself?**
  To answer this question, our study first examine the overall variation structure of Beijing Dialect, then we try to find out whether there is a difference between city and suburban areas of Beijing. Finally we extend our scale of analysis, consider several dialect sites, which are nearby Beijing, together. We want to know that whether Beijing Dialect is effected by the surrounding dialectal varieties.

- **RQ2: Is there a relationship between geographic distance and the phonetic edit distance?**
  We quantitatively measures the phonetic edit distances between dialect sites within Beijing and the nearby area, in order to examine if the geographic distance contributes to explaining the internal dialectal variation.

To address the two research questions, our research based on multi-site dialectal data, computes inter-site distances, and analyzes the results using visualization techniques.

2 Related Work

Nerbonne and Heeringa (1997) proposed a phonetic dialectometry approach based on edit distance, in which the dialect differences are modeled as continuous distance values through the comparison of phonetic transcripts. This established a framework for the dialectometry studies based on the distance measures of strings. With the development of the dialectology, researchers introduced

lots of methods for quantifying and measuring the dialectal differences. However, many early studies were limited in terms of methodological scalability for those who wants to do the modification. For this issue, Shim and Nerbonne (2022) introduced **dialectR**[1], an open-source R package for dialect distance computation, multidimensional scaling, clustering, and visualization in the statistical programming language R.

3 Methodology

Our research broadly follows the analyses pipeline proposed in the dialectR framework. Accordingly, we first prepared the dialectal and geographic data, preprocessed the data, and then continue the distance-based analysis and spatial visualization.

3.1 Dialectal Data

The dialect data we used in the research were obtained from the **Centre for the Protection of Language Resources of China (Centre for the Protection of Language Resources of China, 2023)**, we abbreviate it into "YuBao" in our paper, platform. According to the (Chang et al., 2026), the Chinese Language Resources Protection Project Collection and Display Platform provides parallel (semantically aligned) resources.

During our collection the Beijing Dialect data, we found that the Fangshan (one of the Beijing district) dialect has systematic errors in both transcription and IPA representation, therefore the Fangshan site was excluded from our analysis. The remaining 7 Beijing dialect sites were retained. For each site, the dataset includes unique entries ids, parallel Mandarin vocabularies (which means all the dialect data were based on the same scale of Mandarin vocabularies), IPA pronunciations of each Mandarin vocabulary, and their dialectal transcripts. Each dialect site contains 1,200 lexical entries, we selected the first 300 entries from each site to simplify the computing and analyzing. For the research on the regional influence and differences, we also included dialect sites near Beijing from suburban areas of Hebei Province and Tianjin Municipality. We additionally include the Luanping dialect sites from Hebei Province, whose dialect is regarded as the closest to Standard Mandarin. We used ChatGPT 5.2 for web crawling, because the YuBao platform does not have direct data export.

After obtaining the raw data, we applied several preprocessing steps. First, some entries on the platform were missing. We recorded IDs of the original entries that are missing and removed the corresponding entries from all dialect sites, kept the original ids from the remained entries. Second, in cases where a single Mandarin entries corresponded to multiple expressions within one entries, only the first expression was retained to simplify computation. Some unexpected symbols in the transcript were also removed. Finally, in the original data, the tonal annotations such as "55" could be incorrectly segmented into separate characters "5" and "5" during edit-distance computation. To avoid this issue, we adopted IPA preprocessing procedures from the YuBao project's public GitHub[2] repository. After preprocessing, all cleaned data were merged together for analysis.

3.2 Geographic Data

To map the dialectal distance to geographic space, we first collected latitude and longitude coordinates for each dialect site from the YuBao platform. Then, we collected the administrative boundary data for Beijing, Tianjin, and Hebei from the **GADM**[3] database, combined together with the coordinates data, in order to show the geographic features of all sites. Geographic data were processed using **QGIS**[4] platform and converted into compatible formats for dialectR.

3.3 Edit Distance Computation and Visualization

We first installed and tested the dialectR package. The edit distances between different dialect site-pairs in dialectR are computed based on the processed IPA transcripts. Then we combined the dialectal distance measures with the geographic information and get the spatial visualization in dialectR. During the visualization process, we made modifications to the program to ensure successful visualization. We have three different patterns for the visualization:

- **Mapping the data as the MDS (multidimensional scaling) plot**
  we based on MDS (multidimensional scaling) , in which three dimensions are mapped onto

[1] https://github.com/b05102139/dialectR

[2] https://github.com/kalvinchang/yubao?tab=readme-ov-file

[3] https://gadm.org/

[4] https://qgis.org/

RGB color channels and evenly mixed to represent dialectal variation. During the modification on the dialectR original methods, we found out that the Voronoi package in the dialectR changes the final output a lot. It allows dialect sites to be expanded into spatial regions, while maps without Voronoi visualization only display the phonetic differences at individual points. In this research, we only have 11 dialect sites in total, using Voronoi diagrams alone may reduce the readability, as regional boundaries may harm the point-level variation. Therefore, we generate two MDS plots (with and without Voronoi).

- **Mapping the data by utilizing the results of hierarchical clustering**
  we visualize the results of hierarchical clustering, highlighting group-level similarities among dialect sites.

3.4 Editing of Visualized Results

To improve the readability of the visualizations, the figures were edited using **Inkscape**[5]. Specifically, the maps were cropped, the Hebei and Tianjin area were colored to distinguish the three regions, all dialect sites were labeled accordingly.

4 Experimental Results

The following Figure 1, Figure 2 and Figure 3 present the results of projecting dialectal distances onto geographic space.

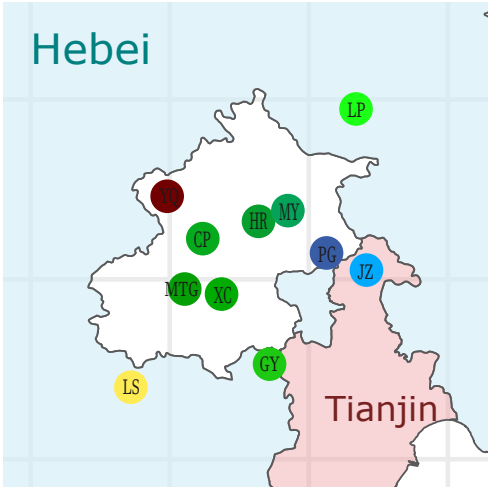


Figure 1: Visualization based on MDS without Voronoi (multidimensional scaling), labeled spots = dialect sites, Beijing: central white area

[5]https://inkscape.org/

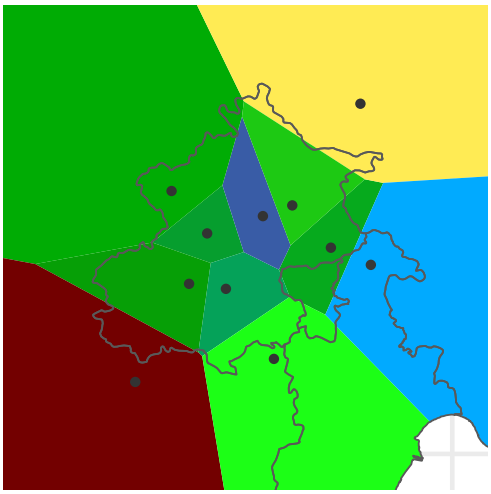


Figure 2: Visualization based on MDS with Voronoi (multidimensional scaling)

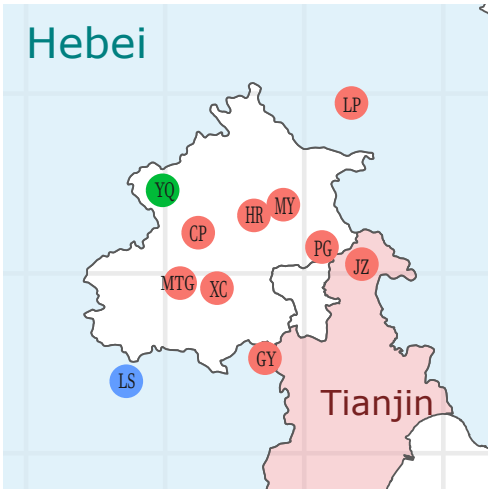


Figure 3: Visualization based on utilizing the results of hierarchical clustering

In Figure 1 and Figure 2, we can see 11 dialects sites across the Beijing, Tianjin and Hebei area. Within the Beijing area, most of the dialect sites show the similar color, but there are also some outliers. As we mentioned before, the MDS plot can map dialectal differences in RGB color, which means that some clear differences can be observed among the dialect sites. As shown in Figure 3, most dialect sites form a single cluster, while only two sites appear as isolated outliers.

Table 1 reports the pairwise edit distances between all dialect sites, computed using dialectR. Based on this full distance matrix, we can get two subsets Table 2 and Table 3 for further analysis. Table 2 presents the pairwise distances among Beijing dialect sites only, illustrating the internal variation within Beijing Dialect. Table 3 summarizes the av-

Table 1: Beijing, Hebei, and Tianjin dialect distance matrix: Dialect Distance Matrix for all sites, CP = Beijing_Changping; HR = Beijing_Huairou; MTG = Beijing_Mentougou; MY = Beijing_Miyun; PG = Beijing_Pinggu; XC = Beijing_Xicheng; YQ = Beijing_Yanqing; LS = Hebei_Laishui; LP = Hebei_Luanping; GY = Hebei_Guangyang; JZ = Tianjin_Jizhou

	CP	HR	MTG	MY	PG	XC	YQ	LS	LP	GY	JZ
CP	.000	.183	.174	.185	.217	.189	.278	.325	.227	.191	.234
HR	.183	.000	.150	.141	.198	.159	.251	.315	.203	.161	.205
MTG	.174	.150	.000	.160	.205	.139	.254	.316	.211	.168	.222
MY	.185	.141	.160	.000	.175	.155	.262	.313	.207	.162	.196
PG	.217	.198	.205	.175	.000	.218	.231	.302	.258	.217	.213
XC	.189	.159	.139	.155	.218	.000	.263	.321	.202	.173	.222
YQ	.278	.251	.254	.262	.231	.263	.000	.320	.304	.265	.298
LS	.325	.315	.316	.313	.302	.321	.320	.000	.308	.294	.339
LP	.227	.203	.211	.207	.258	.202	.304	.308	.000	.179	.237
GY	.191	.161	.168	.162	.217	.173	.265	.294	.179	.000	.221
JZ	.234	.205	.222	.196	.213	.222	.298	.339	.237	.221	.000

rank	site_1	site_2	edit_distance
1	Mentougou	Xicheng	0.139
2	Huairou	Miyun	0.141
3	Huairou	Mentougou	0.150
4	Miyun	Xicheng	0.155
5	Huairou	Xicheng	0.159

Table 2: Pairwise edit distances between Beijing dialect sites (top5)

type	reference	comparison	mean_dist
within_BJ	BJ_7_sites	BJ_7_sites	.199
outside_vs_BJ	HB_LF_GY	BJ_7_sites	.203
outside_vs_BJ	HB_CD_LP	BJ_7_sites	.238

Table 3: Mean edit distances between Beijing sites and selected external reference sites: BJ = Beijing, GY = Guangyang, LP = Luanping

erage internal distance among Beijing dialect sites and compares it with the average edit distance between Beijing dialects and the Luanping dialect, which is commonly regarded as being closest to Standard Mandarin.

Together, these results provide a quantitative basis for comparing internal Beijing dialect variation with variation between Beijing and nearby dialects sites.

5 Result Analysis

Analysis of Figure 1 and Table 3 together shows that there is a clear difference between Beijing Dialect and the Luanping Dialect, and the average edit distance between Luanping and the Beijing sites is higher than the average distance within Beijing Dialect itself. This indicates that although Beijing Dialect and the Luanping Dialect both belong to Northern Mandarin and share overall similarities, they still show clear differences. Variation within Beijing is smaller than the variation between Beijing and Luanping, suggesting a more tightly connected internal structure within Beijing Dialect itself.

By examining Figure 1 together with Table 2, it can be observed that Beijing Dialect is not completely homogeneous and still shows internal variation. In general, dialect sites closer to the city center of Beijing (Xicheng) tend to show smaller differences, while some suburban sites (Yanqing) exhibit larger differences. Specifically, the Yanqing site (YQ) ap-

pears relatively isolated in both the MDS plot and the clustering results. This pattern may be related to Yanqing's location and transportation conditions in mountainous areas.

When we extend the scale of our analysis to nearby areas, Beijing Dialect is found to be more similar to the Guangyang Dialect from Langfang, Hebei. This result is reflected both in Figures and Tables: Guangyang shows values close to Beijing sites in the MDS plot (similar green). This site is also grouped with most of the Beijing sites in the clustering analysis. Among the other Hebei and Tianjin sites, no strong similarities are observed. Although Tianjin Jixian and Beijing Pinggu sites appear similar in color in the MDS plot, the average edit distance in Table 1 shows that their actual dialectal distance remains relatively large.

Combining all the Figures and Tables together, we can say that Beijing Dialect has large overall similarities, but still exhibiting systematic internal spatial variation.

Figure 1 and Table 2 show that the dialectal between the Beijing Mentougou–Xicheng pair and the Beijing Huairou–Miyun pair are very small. Through the map, we can also see that the geographic distances between these pairs are relatively short. This may suggests that geographic distance is in some degree related to dialectal distance.

6 Discussion and Conclusion

As we mentioned before, in order to simplify the computational process, this study retained only the first dialectal realization (IPA and transcript)for each vocabulary. But in practice, dialects often exhibit multiple realizations for the same word, and dialectR also provides methods to account for such variation. In addition, only the first 300 vocabularies were used in our analysis, whereas the actual number of available entries is much larger. When we take multiple expressions and a larger dataset into account, it may lead to substantially different results, which should be explored in the future research. Future research could also integrate measures of geographic distance and topographic features into the analysis, which may lead to a more comprehensive understanding of dialectal variation.

Based on the analyses presented above, it can be observed that the dialectal differences exist both between dialects and among variants within the same dialect. These differences cannot be explained solely by geographic factors, but are more likely the result of the interaction of multiple factors, such as economic development, transportation, education and population. Therefore, variation within and between dialects should not be treated as the linear separable data.

Ethical Statement

ChatGPT 5.2 used in data crawling for the dialects, code for preprocessing(merging the files together and get the first phonetic representation back). Modifying the R code in the dialectR (assistant by data execution).

Acknowledgement

The design of this MaiNLP lab style sheet is strongly inspired by the NERT Overleaf template and adapted for the seminar **NLP for Dialects** at LMU.

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